Hanoi University of Science and Technology
School of Information and Communication Technology

MUSIC TIMBRE TRANSFER USING LATENT DIFFUSION AND LORA FINE-TUNING

PROJECT 2

Student: Vo Ta Quang Nhat - 20225454

Supervisor: Dr. Nguyen Duc Anh

Contents

Abstract

This report presents an experimental system for music timbre transfer based on latent diffusion and Low-Rank Adaptation (LoRA). The goal is to transform a short input audio clip into a target instrument timbre while preserving its melody and rhythmic structure. The system currently supports three target timbres: piano, flute, and guitar. Audio clips are represented as 512×512 mel spectrogram images, which allows the task to be formulated as image-to-image generation in the spectrogram domain. Instead of training on paired source–target examples, I fine-tune one LoRA adapter for each target timbre using unpaired spectrogram images from that timbre. A pretrained Riffusion-style latent diffusion model provides the base spectrogram generation capability, while each LoRA adapter specializes the UNet denoiser toward a particular timbre distribution. During inference, an input waveform is converted into a spectrogram image, transformed through the image-to-image diffusion pipeline, and reconstructed into audio using Griffin–Lim. The system is evaluated using paired test-set spectrogram comparison and an external baseline based on the ISMIR 2024 latent-diffusion model by Demerlé et al. [6]. A demo page is available at this link.

1 Introduction

Music timbre transfer aims to change the perceived instrument identity of an audio signal while keeping its musical content recognizable. In this project, the target task is to take a short input audio clip and generate an output clip with the same melody and rhythm, but with a different instrument timbre. This is a difficult problem because timbre is not an isolated property. It is entangled with pitch, dynamics, articulation, temporal envelope, harmonic structure, and recording conditions. A direct supervised formulation would require aligned performances of the same musical phrase in multiple instruments, which is expensive and difficult to obtain at scale.

My approach uses spectrogram-domain image-to-image generation. First, a short waveform is converted into a mel spectrogram and then represented as an RGB image. A pretrained Riffusion-style latent diffusion model is used as the base generative model. Instead of fully fine-tuning the model, I train separate LoRA adapters for piano, flute, and guitar. Each adapter learns the visual distribution of spectrograms for a single target timbre. At inference time, the input spectrogram is used as the initial image, and the target-timbre LoRA adapter guides the diffusion process toward the desired instrument style.

This direction was chosen after considering several alternatives, including convolutional autoencoders, conditional VAEs, and VQ-VAE-based methods. Those approaches are conceptually suitable for audio representation learning, but they require more data, more training time, and more computational resources than were available for this project. Latent diffusion with LoRA provides a practical compromise: it reuses a strong pretrained model, trains only a small number of additional parameters, and can be trained with accessible GPU resources such as Google Colab.

The rest of this report describes the problem formulation, data representation, latent diffusion model, LoRA fine-tuning strategy, implementation details, evaluation design, experimental results, and limitations of the current system.

2 Problem Definition

Let x be a short source audio waveform and let s be a target timbre label, where:

$$s \in \{\text{piano}, \text{flute}, \text{guitar}\}. \tag{1}$$

The desired output is a waveform $\hat{x}_s$ such that:

- $\hat{x}_s$ preserves the melody, rhythm, and approximate temporal structure of x;
- $\hat{x}_s$ sounds closer to the target timbre s;
- $\hat{x}_s$ avoids severe artifacts such as random notes, excessive noise, unstable pitch, or broken temporal continuity.

In this project, audio is represented through mel spectrograms. The full transformation is expressed as:

$$x \xrightarrow{\text{mel spectrogram}} I_x \xrightarrow{\text{img2img diffusion + LoRA}_s} \hat{I}_s \xrightarrow{\text{spectrogram inversion}} \hat{x}_s, \tag{2}$$

where I_x is the input spectrogram image and $\hat{I}_s$ is the generated target-timbre spectrogram image.

3 Data Representation

3.1 Source Data and Splits

The project uses short timbre-specific spectrogram segments derived from the MAESTRO dataset. The final dataset is organized into three subsets: training, validation, and test. The training subset is used for LoRA optimization, the validation subset is used to monitor training behavior, and the test subset is reserved for inference and evaluation.

This report focuses on the spectrogram representation and metadata organization. The intermediate audio rendering procedure is not discussed in detail because the training pipeline consumes only spectrogram arrays and metadata.

3.2 Spectrogram Representation

Each audio segment is converted into a mel spectrogram array with spatial size 512×512. This array is then converted into an RGB image compatible with the pretrained image diffusion pipeline. The image tensor is normalized to the range $[-1, 1]$ before being passed into the VAE encoder.

Table 1: Spectrogram conversion configuration. Fill or update values if the final code changes.

Parameter	**Value**
Sample rate	44 100 Hz
Image width	512
Number of mel bins	512
FFT size	17640
Hop length	441
Window length	4410
Spectrogram file format	NumPy array `.npy`
Image format for diffusion	RGB image, normalized to $[-1, 1]$

The metadata file stores the following columns:

$$\texttt{maestro\_id}, \quad \texttt{slice}, \quad \texttt{timbres}, \quad \texttt{file\_name}, \quad \texttt{subset}. \tag{3}$$

This metadata allows filtering by subset and timbre, and enables paired evaluation by matching examples with the same `maestro_id` and `slice` but different timbres.

3.3 Dataset Flow

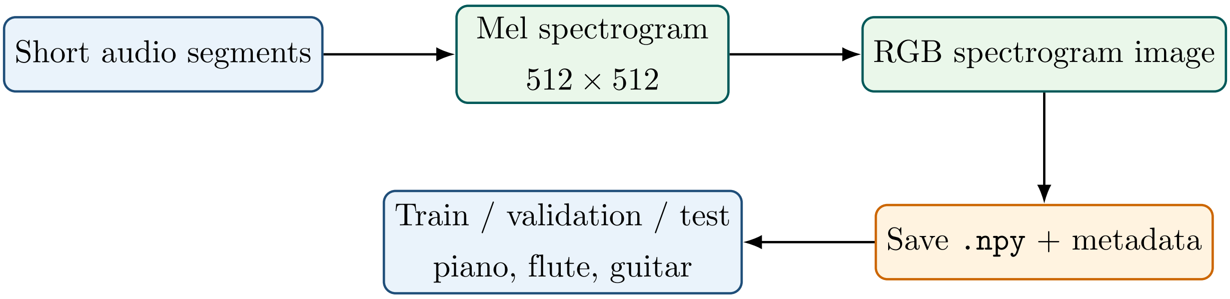


Figure 1: Dataset representation flow. The report focuses on the spectrogram representation and metadata organization rather than the audio rendering procedure.

3.4 Dataset Statistics

Table 2 should be filled after generating the final dataset.

Table 2: Dataset statistics. Replace the placeholder values with the final dataset counts.

Timbre	**Train**	**Validation**	**Test**	**Total**
Piano	800	100	100	1000
Flute	800	100	100	1000
Guitar	800	100	100	1000
Total	2400	1000	1000	10000

4 Model and Method

4.1 Overall Approach

The proposed method treats music timbre transfer as spectrogram image style transfer. A pre-trained latent diffusion model provides the base image generation capability, and LoRA adapters specialize this model toward target instrument timbres. The full system consists of three main stages: LoRA training, image-to-image inference, and waveform reconstruction.

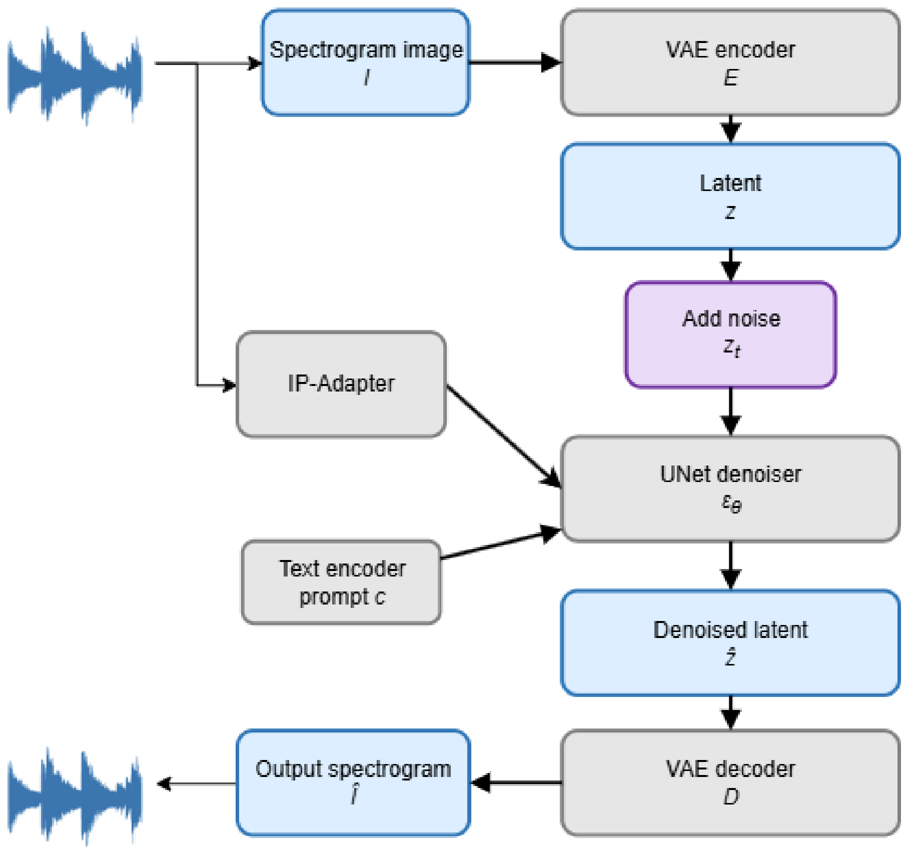


Figure 2: High-level architecture of the proposed spectrogram-domain timbre transfer system. Replace this placeholder with the final architecture diagram if available.

At inference time, I use the image-to-image pipeline from the Hugging Face Diffusers library. The input spectrogram is encoded into latent space, partially noised according to a strength value, and then denoised under the target-timbre prompt and the selected LoRA adapter.

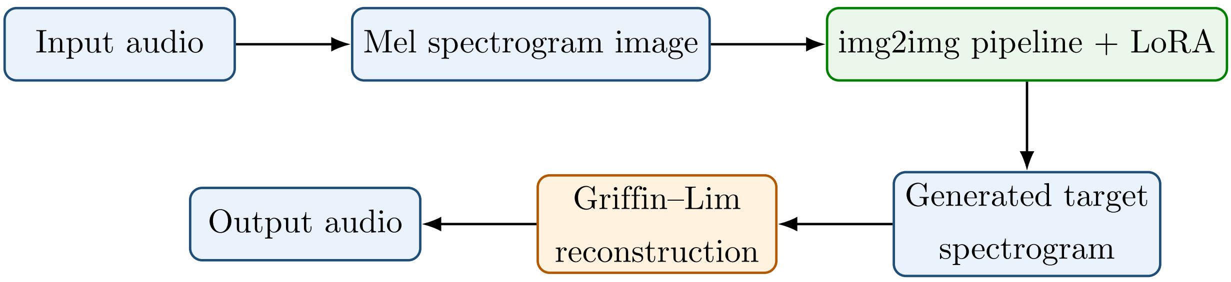


Figure 3: Inference pipeline for timbre transfer using image-to-image diffusion over spectrogram images.

4.2 Stable Diffusion as a Latent Diffusion Model

Originally, I tried CVAE and VQ-VAE as the based model, but they require more computational resource and power than I can provide. So I choosed another approach with latent diffusion model, more specific, the stable diffusion because its easiness for accessibility.

Stable Diffusion is a latent diffusion model. Instead of running diffusion directly in high-dimensional image pixel space, it first compresses an image into a lower-dimensional latent representation using a variational autoencoder (VAE). In the spectrogram setting, the input image

$I \in \mathbb{R}^{3\times512\times512}$ is encoded into a latent tensor $z \in \mathbb{R}^{4\times64\times64}$:

$$z = \mathcal{E}_{\mathrm{VAE}}(I). \tag{4}$$

Diffusion training adds Gaussian noise to the latent according to a timestep t. The forward noising process can be written as:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \tag{5}$$

where z_0 is the clean latent and z_t is the noisy latent at timestep t. The UNet denoiser receives z_t, the timestep t, and a text-conditioning vector c, then predicts the noise component:

$$\hat{\epsilon}_\theta = \epsilon_\theta(z_t, t, c). \tag{6}$$

For this project, the conditioning text is a short prompt such as:

"a mel spectrogram of solo flute music".

During image-to-image inference, the source spectrogram is not generated from pure noise. Instead, it is encoded into latent space and only partially noised. This makes the denoising strength a key control parameter: a low strength preserves more source structure, while a high strength allows stronger timbre transformation but increases the risk of changing notes, rhythm, or temporal continuity.

4.3 Low-Rank Adaptation

LoRA is used to adapt the diffusion model without updating the full model parameters. For a frozen linear weight matrix $W \in \mathbb{R}^{d\times k}$, LoRA adds a low-rank trainable update:

$$W' = W + \Delta W, \tag{7}$$

where:

$$\Delta W = \frac{\alpha}{r} BA, \tag{8}$$

with $A \in \mathbb{R}^{r\times k}$, $B \in \mathbb{R}^{d\times r}$, rank r, and scaling factor α. Since r is much smaller than d and k, the number of trainable parameters is much smaller than full fine-tuning. This makes LoRA suitable for limited data and limited GPU memory.

In this project, LoRA adapters are inserted into attention projection modules inside the UNet:

$$\texttt{to\_q}, \quad \texttt{to\_k}, \quad \texttt{to\_v}, \quad \texttt{to\_out.0}. \tag{9}$$

These modules control the query, key, value, and output projections in attention layers. Updating them allows the model to alter how spectrogram structure attends to text conditioning and latent features, while the original pretrained model remains frozen.

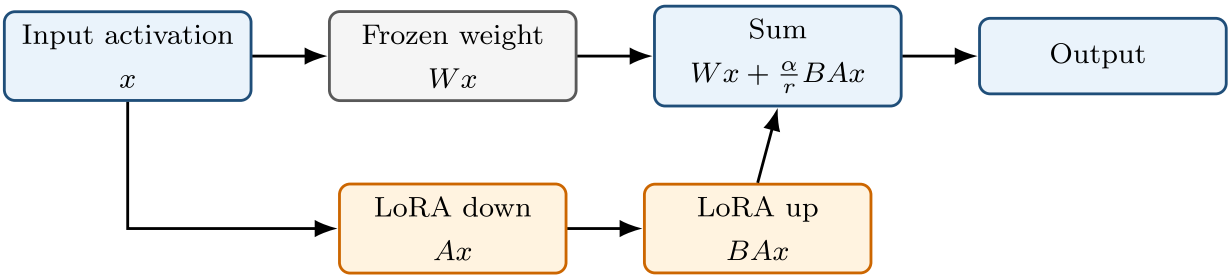


Figure 4: LoRA adds a trainable low-rank update branch in parallel to a frozen weight matrix.

4.4 LoRA Training Strategy

The system does not train a direct paired mapping such as piano-to-flute or flute-to-guitar. Instead, it trains one independent LoRA adapter for each target timbre:

$$\text{Base diffusion model} + \text{LoRA}_{\text{piano}}, \tag{10}$$

$$\text{Base diffusion model} + \text{LoRA}_{\text{flute}}, \tag{11}$$

$$\text{Base diffusion model} + \text{LoRA}_{\text{guitar}}. \tag{12}$$

During piano LoRA training, the model only sees piano spectrogram images. The same principle is applied to flute and guitar. Therefore, each adapter learns a target timbre distribution rather than a deterministic source-to-target conversion rule.

This strategy is simple and modular. Adding a new instrument only requires training a new adapter, while the base model and inference pipeline remain unchanged. The limitation is that content preservation is not learned explicitly from paired examples. Instead, it depends mainly on the image-to-image initialization and the denoising strength parameter.

4.5 Diffusion Training Objective

The LoRA adapter is trained using the diffusion noise-prediction objective. Given an input spectrogram image I, the VAE encoder maps it into a latent representation z. A timestep t and Gaussian noise ϵ are sampled, and the scheduler creates a noisy latent z_t. The UNet predicts the added noise conditioned on the timestep and text prompt:

$$\hat{\epsilon}_\theta = \epsilon_\theta(z_t, t, c), \tag{13}$$

where c is the text embedding of the prompt, such as *"a mel spectrogram of solo piano music"*. The training loss is:

$$\mathcal{L}_{\text{diffusion}} = \mathbb{E}_{z,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2\right]. \tag{14}$$

Only the LoRA parameters are updated. The VAE, text encoder, and original UNet weights remain frozen.

5 Implementation

5.1 Training Configuration

Table 3 records the main training hyperparameters. Fill in the final values after running experiments.

Table 3: Training configuration.

Hyperparameter	**Value**
Base checkpoint	`riffusion/riffusion-model-v1`
Image size	512
LoRA rank	16
LoRA alpha	16
Batch size	1
Gradient accumulation	4
Effective batch size	4
Learning rate	1×10^{-4}
Max training steps	5000
Mixed precision	fp16
Gradient checkpointing	Enabled
Validation interval	500 steps
Validation max batches	8
Max validation samples	0
Checkpoint interval	500 steps
Checkpoint limit	4
Dataloader workers	2
Prompt template	`a mel spectrogram of solo {timbre} music`

5.2 Inference Configuration

The main inference parameters are image-to-image strength, classifier-free guidance scale, and the number of denoising steps. Strength is particularly important because it controls the balance between content preservation and timbre transformation.

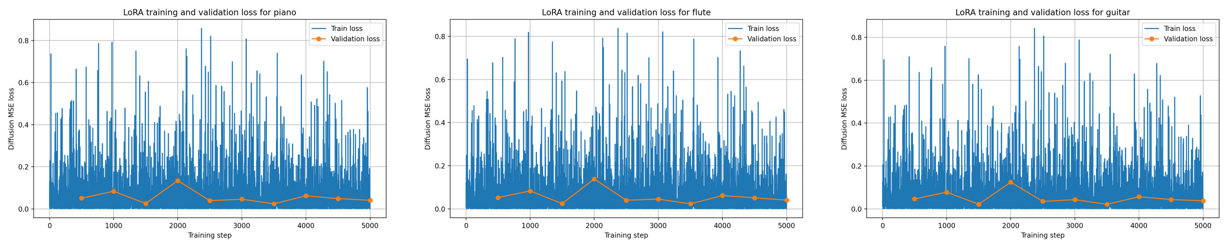
Table 4: Inference parameters and expected effects.

Parameter	**Typical value**	**Expected effect**
Strength	0.20–0.40	Higher values produce stronger timbre change but may damage melody/rh
Guidance scale	2.0–4.0	Higher values follow the prompt more strongly but may create artifacts
Inference steps	30–50	More steps may improve detail but increases runtime

6 Evaluation Design

6.1 Training and Validation Loss

During training, the diffusion mean squared error loss is recorded for both training and validation. Training loss is logged every optimization step, while validation loss is computed periodically on held-out spectrograms from the same target timbre.



(a) Piano LoRA (b) Flute LoRA (c) Guitar LoRA

Figure 5: Training/validation curves. Fill these images using the saved `metrics.csv` files after training.

In preliminary experiments, the loss curves were not always smooth or strictly decreasing. This behavior is common in small-scale LoRA experiments for generative models, where validation loss does not always correlate perfectly with perceptual quality. Therefore, the final evaluation also includes spectrogram comparison and listening-based qualitative inspection.

6.2 Paired Test-Set Evaluation

Although training is unpaired, the synthetic dataset can support paired evaluation. Since different timbre spectrograms share the same `maestro_id` and `slice`, a generated output can be compared to a target-timbre spectrogram with the matching identity and slice index. For example:

$$I_{\text{flute}}^{(m,k)} \to \hat{I}_{\text{piano}}^{(m,k)} \quad \text{compared with} \quad I_{\text{piano}}^{(m,k)}, \tag{15}$$

where m is the file index and k is the slice index.

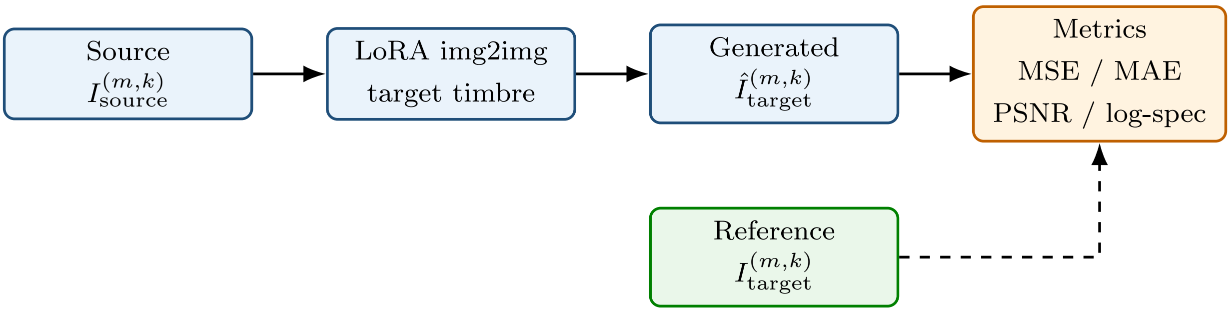


Figure 6: Paired evaluation using matching `maestro_id` and slice index (m, k).

6.3 External Reference Evaluation

A second evaluation mode compares the LoRA output with an output generated by an external model, which achieve very good results. The external baseline used in this report is the latent-diffusion method from *Combining Audio Control and Style Transfer Using Latent Diffusion* by Demerlé et al. [6]. That work aims to separate local musical structure from global timbre/style information, enabling audio generation that matches a timbre target while preserving structure from another source.

This reference should not be interpreted as absolute ground truth. Instead, it serves as a comparison against a dedicated audio-domain timbre-transfer model. This is useful because the proposed method operates through image-to-image generation over spectrograms, whereas the external baseline is designed directly for audio style transfer.

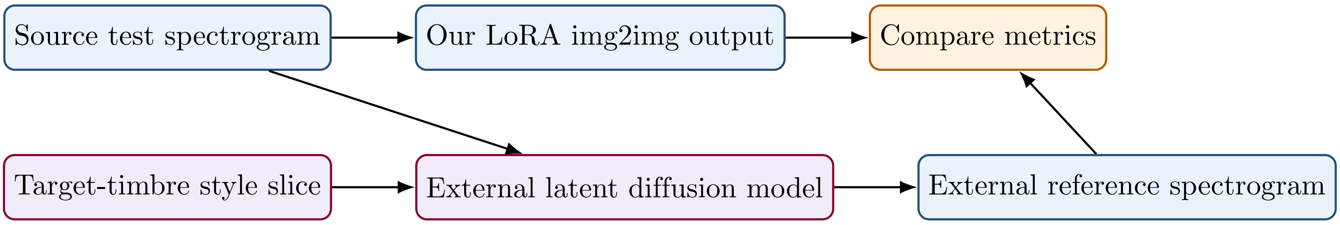


Figure 7: External reference evaluation using the latent-diffusion timbre-transfer baseline from Demerlé et al. [6].

6.4 Metrics

The evaluation script records the following metrics:

Table 5: Evaluation metrics.

Metric	**Direction**	**Meaning**
Image MSE	Lower is better	MSE between generated and reference spectrogram images
Image MAE	Lower is better	Pixel-level absolute difference
Image PSNR	Higher is better	Image similarity score derived from MSE
Log-spectrogram MSE	Lower is better	Squared error between log-scaled spectrogram magnitudes
Log-spectrogram MAE	Lower is better	Absolute error between log-scaled spectrogram magnitudes

These metrics are useful for quantitative comparison, but they do not fully capture perceptual audio quality. Therefore, qualitative listening and visual spectrogram inspection are also included.

7 Evaluation Results

For evaluation, this is the hard part for generation problem. I do two type of evaluation:

1. create systhesis audio from Maestro dataset MIDI file, and compare it to the result by the mode

2. compare to better model, I choose the one which is from ISMIS2024 [6]

Table 6: Paired test-set evaluation against the corresponding target-timbre spectrogram.

Direction	**Image MSE**	**Image MAE**	**PSNR**	**LogSpec MSE**	**LogSpec MAE**
Piano → Flute	0.000415	0.017918	33.980591	4.980396	2.060197
Piano → Guitar	0.000245	0.013269	36.284186	3.291724	1.615721
Flute → Piano	0.000199	0.011393	37.820174	2.840601	1.440390
Flute → Guitar	0.000172	0.010448	38.246726	2.249579	1.260138
Guitar → Piano	0.000276	0.014483	35.797980	3.979638	1.820110
Guitar → Flute	0.000461	0.018722	33.646075	5.383339	2.133092

Table 7: Comparison with external model baseline [6].

Direction	Image MSE	Image MAE	PSNR	LogSpec MSE	LogSpec MAE
Piano → Flute	0.000157	0.009232	38.066780	1.424865	0.970212
Piano → Guitar	0.000166	0.009613	37.856165	1.351293	0.937687
Flute → Piano	0.000160	0.009245	38.045073	1.265074	0.894014
Flute → Guitar	0.000150	0.008905	38.337503	1.252227	0.885770
Guitar → Piano	0.000179	0.009888	37.582137	1.314197	0.921626
Guitar → Flute	0.000162	0.009514	37.952748	1.681170	1.076653

The external baseline obtains lower error values than the proposed LoRA-based pipeline, which is expected because it is designed specifically for audio control and style transfer. However, the LoRA approach remains useful as a lightweight and modular baseline. It requires only target-timbre spectrogram examples, trains a small adapter for each timbre, and can be extended to new instruments without retraining the full diffusion model.

8 Discussion

8.1 Effectiveness of LoRA Training

The main advantage of the LoRA setup is that it avoids paired source-target training data. Each adapter only needs examples from the target timbre distribution. This makes the training process simple and modular: adding a new instrument such as guitar requires training a new adapter rather than redesigning the full model. The drawback is that the model is not explicitly optimized to preserve melody during timbre conversion. Content preservation mostly comes from the image-to-image initialization and the denoising strength.

The approach is also computationally practical. Because only LoRA parameters are trained, the experiments can be run using limited resources such as the free-tier Colab T4 GPU. This makes the method suitable as a feasible this project baseline.

8.2 Trade-off between Content Preservation and Timbre Change

The denoising strength is the most important inference parameter. Lower strength values tend to preserve the input spectrogram structure, which helps maintain melody and rhythm, but the timbre change may be weak. Higher strength values push the image more strongly toward the target timbre distribution, but they may alter the note structure or introduce artifacts.

After multiple trials, I selected a strength of 0.5 and a guidance scale of 3.0 as a practical balance for the current model. This setting produced a clearer timbre change while still preserving the note structure reasonably well. However, the best value may depend on the target timbre, the selected checkpoint, and the input audio.

8.3 Limitations of Spectrogram Reconstruction

The current pipeline reconstructs audio from generated spectrograms using Griffin–Lim. This is simple and does not require training an additional vocoder, but it can introduce artifacts and may reduce perceptual quality. Therefore, poor audio quality may come from either the diffusion output or the spectrogram inversion step. Future work should separate these factors more carefully, possibly by using a stronger neural vocoder.

8.4 Metric Limitations

Image-level metrics such as MSE, MAE, and PSNR provide useful quantitative summaries, but they do not fully reflect musical quality. Two spectrograms can be close in pixel space while sounding different, or visually different while preserving the same melody. For this reason, this project reports both quantitative metrics and qualitative examples.

9 Future Work

Possible future improvements include:

- experimenting with Constant-Q Transform (CQT) representations in addition to mel spectrograms;
- replacing Griffin–Lim with a neural vocoder for higher-quality waveform reconstruction;
- adding overlap-add inference for longer audio clips;
- tuning LoRA rank, LoRA weight, guidance scale, and denoising strength separately for each instrument;
- adding perceptual audio metrics and human listening evaluation.

10 Conclusion

This project investigates unpaired music timbre transfer using LoRA fine-tuning of a diffusion-based spectrogram model. The system converts audio into mel spectrogram images, adapts a pretrained Riffusion-style latent diffusion model using target-specific LoRA adapters, and reconstructs generated spectrograms back into audio. The current setup supports piano, flute, and guitar. The approach is modular and data-efficient because each target timbre is trained independently using unpaired examples. Evaluation combines training/validation loss, paired test-set comparison, external baseline comparison, and qualitative audio/spectrogram inspection. Although the current system is still limited by content preservation, checkpoint selection, and Griffin-Lim reconstruction quality, it provides a feasible baseline for diffusion-based timbre transfer under limited computational resources.

I would like to thank my supervisor, Dr. Nguyen Duc Anh, for his support and guidance throughout this project. His feedback helped me identify weaknesses in early approaches and improve the final direction of the system.

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